Fahim Faisal

LLM Post-Training, Evaluation & Agentic Systems

ffaisal.gmu@gmail.com ▪ +1 571-536-1901 ▪ fahimfaisal.info ▪ GitHub ▪ LinkedIn ▪ Google Scholar

Knoxville, Tennessee, USA ▪ **U.S. Permanent Resident** — authorized to work without sponsorship

EDUCATION

Ph.D., Computer Science — Fairfax, Virginia, USA, January 2020 – May 2026

George Mason University | GPA 4.00/4.00

M.S., Computer Science — Fairfax, Virginia, USA, January 2020 – May 2023

George Mason University | GPA 4.00/4.00

B.Sc., Computer Science and Engineering — Gazipur, Bangladesh, December 2012 – November 2016

Islamic University of Technology | GPA 3.78/4.00

TECHNICAL SKILLS

Post-Training	RL post-training (PPO, GRPO, RLVR, REINFORCE++), preference optimization (DPO, KTO), parameter-efficient fine-tuning (LoRA/adapters), knowledge distillation, synthetic data generation
Evaluation	`lm-evaluation-harness`, custom benchmark design, calibration & robustness analysis, automated LLM-as-a-judge pipelines, statistical significance testing
Agentic & RAG	Multimodal agentic RAG, retrieval pipelines, tool-use augmentation, multi-step reasoning over heterogeneous data
ML Stack	Python, PyTorch, Hugging Face (Transformers/TRL/PEFT), vLLM, DeepSpeed, CUDA, Weights & Biases
Infrastructure	Slurm multi-GPU workflows, Docker, distributed training & inference, Grafana monitoring, HPC clusters
Data	Spark, SQL, pandas, NumPy, Arrow; large-scale ETL and graph-structured data pipelines

EXPERIENCE

Postdoctoral Research Associate — Oak Ridge, Tennessee, USA, July 2026 – Present

Oak Ridge National Laboratory

- Build multimodal agentic RAG systems for scientific discovery, combining retrieval, tool use, and multi-step reasoning over heterogeneous scientific data.
- Adapt pretrained single-cell foundation models to toxicological response prediction, bridging toxicogenomics and single-cell representation learning.

GenAI Research Intern — San Jose, California, USA, Summer 2025

Zoom Video Communications

- Built a multilingual reasoning alignment pipeline distilling verifiable semantics from a high-resource expert model into smaller multilingual models; ran RL post-training (GRPO, PPO, REINFORCE++) across 20+ languages, 5+ model variants, and multiple reward functions (paper).
- Stood up benchmark evaluation with `lm-evaluation-harness` across 100+ language varieties, adding calibration and robustness analysis; containerized the evaluation and inference stack with Docker for reproducible runs.

Applied Research Intern (Ph.D.) — California, USA, Summer 2024

eBay Inc.

- Designed safety and policy alignment on top of eBay's in-house e-commerce LLM (LiLiuM).

- Generated 1M+ synthetic preference examples using multiple LLM families (Phi, LLaMA, GPT) to build the alignment training set.
- Built an automated compliance evaluation quantifying refusal quality against false-positive rates, giving the team a repeatable measure of the over-refusal tradeoff.

Applied Research Intern (Ph.D.) California, USA
Summer 2023

eBay Inc.

- Implemented the Pinterest `Pixie` random-walk algorithm for aspect-based recommendation over `user-item-aspect bipartite graphs`, trained on 1.3M interactions across 3 major product categories.
- Improved offline ranking metrics (NDCG, MRR, Recall@K) over the existing baseline; designed the supporting Spark/SQL feature and training pipelines.

Graduate Research Assistant Fairfax, Virginia, USA
June 2020 – May 2025

George Mason NLP Lab

- Designed and shipped DialectBench, a reproducible evaluation suite spanning **100+ language varieties** across reasoning, toxicity, QA, and machine translation — ACL 2024 Best Social Impact Paper.
- Trained and evaluated models from BERT-scale encoders to large dense LLMs on institutional multi-GPU Slurm clusters; deployed and served models for internal evaluation.
- Ran statistical analysis of model behavior including calibration evaluation and significance testing.

Software Engineer Dhaka, Bangladesh
March 2017 – September 2017

Samsung R&D Institute

- Developed system-level components for the Samsung Tizen wearable OS.

AWARDS

- **Best Social Impact Paper Award**, ACL 2024
- **Outstanding PhD Student Award**, George Mason University CS, 2024

PUBLICATIONS

2025

Faisal, F., Song, K., Wang, S., Ma, S., Liu, S., Deng, H., & Indurthi, S. R. (2025). *Aligning multilingual reasoning with verifiable semantics from a high-resource expert model.* Retrieved from https://arxiv.org/abs/2509.25543 (Under Review at ARR)

Faisal, F., Rahman, M. M., & Anastasopoulos, A. (2025, November). Dialectal toxicity detection: Evaluating LLM-as-a-judge consistency across language varieties. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Findings of the association for computational linguistics: Emnlp 2025* (pp. 12429–12452). Suzhou, China: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2025.findings-emnlp.664/ doi: 10.18653/v1/2025.findings-emnlp.664

Faisal, F., & Anastasopoulos, A. (2025, January). Testing the boundaries of LLMs: Dialectal and language-variety tasks. In Y. Scherrer, T. Jauhiainen, N. Ljubešić, P. Nakov, J. Tiedemann, & M. Zampieri (Eds.), *Proceedings of the 12th workshop on nlp for similar languages, varieties and dialects* (pp. 68–92). Abu Dhabi, UAE: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2025.vardial-1.6/

2024

Faisal, F., Ahia, O., Srivastava, A., Ahuja, K., Chiang, D., Tsvetkov, Y., & Anastasopoulos, A. (2024, August). DIALECTBENCH: An NLP benchmark for dialects, varieties, and closely-related languages. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 14412–14454). Bangkok, Thailand: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2024.acl-long.777 doi: 10.18653/v1/2024.acl-long.777 **[Best Social Impact Paper Award]**

Faisal, F., & Anastasopoulos, A. (2024b, November). An efficient approach for studying cross-lingual transfer in multilingual language models. In J. Sälevä & A. Owodunni (Eds.), *Proceedings of the fourth workshop on multilingual representation learning (mrl 2024)* (pp. 45–92). Miami, Florida, USA: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2024.mrl-1.4

Faisal, F., & Anastasopoulos, A. (2024a, June). Data-augmentation-based dialectal adaptation for LLMs. In Y. Scherrer, T. Jauhiainen, N. Ljubešić, M. Zampieri, P. Nakov, & J. Tiedemann (Eds.), *Proceedings of the eleventh workshop on nlp for similar languages, varieties, and dialects (vardial 2024)* (pp. 197–208). Mexico City, Mexico: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2024.vardial-1.17 doi: 10.18653/v1/2024.vardial-1.17

2023

Faisal, F., & Anastasopoulos, A. (2023, December). Geographic and geopolitical biases of language models. In D. Ataman (Ed.), *Proceedings of the 3rd workshop on multi-lingual representation learning (mrl)* (pp. 139–163). Singapore: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2023.mrl-1.12/ doi: 10.18653/v1/2023.mrl-1.12

Song, Y., Cui, C., Khanuja, S., Liu, P., Faisal, F., Ostapenko, A., ... Neubig, G. (2023). Globalbench: A benchmark for global progress in natural language processing. *Accepted at EMNLP 2023 main conference*. Retrieved from https://arxiv.org/abs/2305.14716

Rahman, M. M., Sakib, F. A., Faisal, F., & Anastasopoulos, A. (2023). To token or not to token: A comparative study of text representations for cross-lingual transfer. *Accepted at Multilingual Representation Learning (MRL) Workshop 2023, Co-located with EMNLP 2023*.

ibn Alam, M. M., Xie, R., Faisal, F., & Anastasopoulos, A. (2023, July). Gmnlp at semeval-2023 tasks 12: Sentiment analysis with phylogeny-based adapters. In *Proceedings of the 17th international workshop on semantic evaluation (semeval-2023)*. Toronto, Canada: Association for Computational Linguistics. Retrieved from https://arxiv.org/abs/2304.12979

2022

Faisal, F., Wang, Y., & Anastasopoulos, A. (2022, May). Dataset geography: Mapping language data to language users. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 3381–3411). Dublin, Ireland: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2022.acl-long.239 doi: 10.18653/v1/2022.acl-long.239

Faisal, F., & Anastasopoulos, A. (2022, November). Phylogeny-inspired adaptation of multilingual models to new languages. In *Proceedings of the 2nd conference of the asia-pacific chapter of the association for computational linguistics and the 12th international joint conference on natural language processing (volume 1: Long papers)* (pp. 434–452). Online only: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2022.aacl-main.34

2021

Faisal, F., Keshava, S., Alam, M. M. I., & Anastasopoulos, A. (2021, November). SD-QA: Spoken dialectal question answering for the real world. In *Findings of the association for computational linguistics: Emnlp 2021* (pp. 3296–3315). Punta Cana, Dominican Republic: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2021.findings-emnlp.281

Faisal, F., & Anastasopoulos, A. (2021, November). Investigating post-pretraining representation alignment for cross-lingual question answering. In *Proceedings of the 3rd workshop on machine reading for question answering* (pp. 133–148). Punta Cana, Dominican Republic: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2021.mrqa-1.14

Mahmud, J., Faisal, F., Arnob, R. I., Anastasopoulos, A., & Moran, K. (2021, August). Code to comment translation: A comparative study on model effectiveness & errors. In *Proceedings of the 1st workshop on natural language processing for programming (nlp4prog 2021)* (pp. 1–16). Online: Association for Computational Linguistics. Retrieved from https://aclanthology.org/2021.nlp4prog-1.1 doi: 10.18653/v1/2021.nlp4prog-1.1

2020

Choudhury, N., Faisal, F., & Khushi, M. (2020). Mining temporal evolution of knowledge graphs and genealogical features for literature-based discovery prediction. *Journal of Informetrics*, *14*(3), 101057. Retrieved from https://www.sciencedirect.com/science/article/pii/S1751157719304468 doi: https://doi.org/10.1016/j.joi.2020.101057

REFERENCES

Dr. Antonios Anastasopoulos
Assistant Professor, Computer Science
George Mason University
antonis@gmu.edu

Dr. Ziyu Yao
Assistant Professor, Computer Science
George Mason University
ziyuyao@gmu.edu

Dr. Yulia Tsvetkov
Associate Professor, Allen School of CSE
University of Washington
yuliats@cs.washington.edu